

Brutal Paper B26 - Light / Soil / Simple Equations / A Peek Beyond the Point

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. An object stands 2.5 m in front of a flat mirror. What is the total distance between the object and its mirror image? [\[Light\]](#)
 - (A) 5.5 m
 - (B) 10 m
 - (C) 0 m
 - (D) 1.25 m
 - (E) 2.5 m
 - (F) 5 m
 - (G) 7.5 m

2. Five times a number, decreased by 8, equals 37. What is the number? [\[Simple Equations\]](#)
 - (A) 5.8
 - (B) 9
 - (C) 45
 - (D) 7
 - (E) 29
 - (F) 8.4
 - (G) 6

3. During a field experiment, a soil sample let 200 mL of water seep right through it in 25 minutes. Find its percolation rate. [\[Soil\]](#)
 - (A) 8 mL/min
 - (B) 12.5 mL/min
 - (C) 10 mL/min
 - (D) 5000 mL/min
 - (E) 0.125 mL/min
 - (F) 16 mL/min

4. First work out $5.6 + 3.45 - 2.8$, then multiply your result by 10. What do you get? [\[A Peek Beyond the Point\]](#)
 - (A) 6.25
 - (B) 625
 - (C) 82.5
 - (D) 52.5
 - (E) 62.5
 - (F) 6.205
 - (G) 60.25

5. A boy is 80 cm from a plane mirror and then walks 30 cm straight toward it. How far apart are he and his image now? [\[Light\]](#)
- (A) 110 cm
 - (B) 100 cm
 - (C) 60 cm
 - (D) 30 cm
 - (E) 50 cm
 - (F) 200 cm
6. Adding 7 to twice a number gives a result that is 4 less than 35. Find the number. [\[Simple Equations\]](#)
- (A) 12
 - (B) 24
 - (C) 21
 - (D) 11
 - (E) 9.5
 - (F) 14
 - (G) 19
7. Sample A lets 250 mL of water through in 20 minutes; sample B lets 250 mL through in 50 minutes. Which soil drains faster, and what is A's percolation rate? [\[Soil\]](#)
- (A) A drains faster; 12.5 mL/min
 - (B) B drains faster; 12.5 mL/min
 - (C) A drains faster; 0.08 mL/min
 - (D) A drains faster; 20 mL/min
 - (E) A drains faster; 5 mL/min
 - (F) B drains faster; 5 mL/min
8. What is the value of 0.6 multiplied by 0.05? [\[A Peek Beyond the Point\]](#)
- (A) 0.3
 - (B) 0.03
 - (C) 3.0
 - (D) 0.11
 - (E) 0.65
 - (F) 0.003
9. Two plane mirrors face each other at an angle of 60 degrees. How many images of a small object placed between them are formed? [\[Light\]](#)
- (A) 5
 - (B) 4
 - (C) 7
 - (D) 3
 - (E) 2
 - (F) 6

10. Three consecutive whole numbers add together to make 72. Which of them is the greatest? [\[Simple](#)

[Equations\]](#)

- (A) 23
- (B) 25
- (C) 27
- (D) 21
- (E) 24
- (F) 22

11. Which soil type has the highest percolation rate and drains fastest because of its large particles? [\[Soil\]](#)

- (A) Both clayey and loamy
- (B) Humus
- (C) All soils equally
- (D) Sandy soil
- (E) Silty soil
- (F) Loamy soil
- (G) Clayey soil

12. Divide 4.5 by 0.9. What is the quotient? [\[A Peek Beyond the Point\]](#)

- (A) 50
- (B) 5
- (C) 0.5
- (D) 4.5
- (E) 4.05
- (F) 0.2
- (G) 45

13. When two plane mirrors are kept at right angles to each other, how many images of an object between them appear? [\[Light\]](#)

- (A) 2
- (B) 5
- (C) 4
- (D) 3
- (E) 90
- (F) 8
- (G) 1

14. Right now a man's age is four times his son's age, and after 5 years it will be three times the son's age then. Find the son's present age. [\[Simple Equations\]](#)

- (A) 5 years
- (B) 20 years
- (C) 15 years
- (D) 12 years
- (E) 30 years
- (F) 40 years
- (G) 10 years

15. Which soil holds the most water and is therefore best for growing paddy (rice)? [\[Soil\]](#)

- (A) Rocky soil
- (B) Sandy soil
- (C) Loamy soil
- (D) Gravel
- (E) Clayey soil
- (F) Pure silt

16. A ribbon 12.5 m long is cut into pieces, each 0.25 m long. How many pieces are obtained? [\[A Peek Beyond the Point\]](#)

- (A) 5
- (B) 50
- (C) 3.125
- (D) 12.75
- (E) 500
- (F) 20
- (G) 25

17. A ray of light strikes a plane mirror making an angle of incidence of 35 degrees. What is the angle between the incoming ray and the reflected ray? [\[Light\]](#)

- (A) 90 degrees
- (B) 140 degrees
- (C) 20 degrees
- (D) 55 degrees
- (E) 35 degrees
- (F) 70 degrees
- (G) 110 degrees

18. The length of a rectangle is 3 cm more than twice its width, and its perimeter is 36 cm. Find the width. [\[Simple Equations\]](#)

- (A) 9 cm
- (B) 10 cm
- (C) 13 cm
- (D) 6 cm
- (E) 8 cm
- (F) 4 cm
- (G) 5 cm

19. A balanced mixture of sand, clay and silt with plenty of humus, ideal for most crops, is called what kind of soil? [\[Soil\]](#)

- (A) Gravel
- (B) Sandy soil
- (C) Loamy soil
- (D) Clayey soil
- (E) Chalky soil
- (F) Peaty soil
- (G) Pure silt

20. Find the value of $7 - 2.35 - 1.8$. [\[A Peek Beyond the Point\]](#)

- (A) 2.85
- (B) 12.85
- (C) 2.95
- (D) 1.85
- (E) 3.85
- (F) 2.15
- (G) 4.85

21. At what angle of incidence must light hit a plane mirror so that the reflected ray is exactly perpendicular to the incident ray? [\[Light\]](#)

- (A) 0 degrees
- (B) 60 degrees
- (C) 90 degrees
- (D) 22.5 degrees
- (E) 45 degrees
- (F) 30 degrees

22. One-third of a number, plus 6, equals 14. What is the number? [\[Simple Equations\]](#)

- (A) 8
- (B) 24
- (C) 18
- (D) 6
- (E) 60
- (F) 20
- (G) 2.67

23. A 500 g dry soil sample soaks up water until it weighs 650 g. What percentage of its dry weight is the water it now holds? [\[Soil\]](#)

- (A) 3%
- (B) 150%
- (C) 13%
- (D) 15%
- (E) 23%
- (F) 43%
- (G) 30%

24. Multiply 2.4 by 1.5, then subtract 0.6 from the product. What is the result? [\[A Peek Beyond the Point\]](#)

- (A) 2.0
- (B) 30
- (C) 36
- (D) 3.0
- (E) 2.4
- (F) 3.6
- (G) 4.2

- 25.** A beam of light makes an angle of 20 degrees with the surface of a plane mirror. What is its angle of reflection, measured from the normal? [\[Light\]](#)
- (A) 70 degrees
 - (B) 35 degrees
 - (C) 110 degrees
 - (D) 90 degrees
 - (E) 20 degrees
 - (F) 160 degrees
- 26.** Riya has some marbles. She gives away 12, then doubles what is left, ending with 36 marbles. How many did she start with? [\[Simple Equations\]](#)
- (A) 42
 - (B) 18
 - (C) 6
 - (D) 48
 - (E) 24
 - (F) 30
 - (G) 36
- 27.** Sandy soil holds 12 mL of water per 100 g and clayey soil holds 45 mL per 100 g. How much more water does 300 g of clayey soil hold than 300 g of sandy soil? [\[Soil\]](#)
- (A) 99 mL
 - (B) 171 mL
 - (C) 36 mL
 - (D) 11 mL
 - (E) 57 mL
 - (F) 33 mL
 - (G) 135 mL
- 28.** In the number 47.069, what is the place value of the digit 6? [\[A Peek Beyond the Point\]](#)
- (A) 0.66
 - (B) 0.6
 - (C) 0.06 (six hundredths)
 - (D) 0.0006
 - (E) 60
 - (F) 6
- 29.** At noon a 2 m vertical pole casts a 3 m shadow. At the same moment a nearby tree casts a 12 m shadow. Using straight-line light, how tall is the tree? [\[Light\]](#)
- (A) 6 m
 - (B) 9 m
 - (C) 24 m
 - (D) 18 m
 - (E) 4 m
 - (F) 16 m
 - (G) 8 m

30. The cost of 5 identical pens equals 90 rupees minus the cost of one such pen. Find the cost of one pen. [\[Simple Equations\]](#)

- (A) 16 rupees
- (B) 12 rupees
- (C) 15 rupees
- (D) 10 rupees
- (E) 75 rupees
- (F) 18 rupees
- (G) 20 rupees

31. When 60 mL of water is poured onto a soil and 18 mL drains out the bottom, what fraction of the water did the soil retain? [\[Soil\]](#)

- (A) 0.42
- (B) 1.4
- (C) 0.7
- (D) 0.3
- (E) 0.25
- (F) 0.6
- (G) 0.18

32. Write the fraction three-eighths, that is $\frac{3}{8}$, as a decimal number. [\[A Peek Beyond the Point\]](#)

- (A) 0.38
- (B) 0.125
- (C) 0.275
- (D) 3.8
- (E) 0.835
- (F) 0.375
- (G) 2.667

33. A dentist uses one mirror to see an enlarged image of a tooth, while a car's side mirror gives a smaller, wider view. Which two mirrors are these, in that order? [\[Light\]](#)

- (A) Concave for the dentist, plane for the car
- (B) Concave for the dentist, convex for the car
- (C) Concave for both
- (D) Both are plane mirrors
- (E) Convex for both
- (F) Convex for the dentist, concave for the car
- (G) Plane for the dentist, convex for the car

34. A number is as much above 50 as it is below 80. Find the number. [\[Simple Equations\]](#)

- (A) 60
- (B) 75
- (C) 130
- (D) 30
- (E) 15
- (F) 65
- (G) 100

35. Over very long periods, soil is formed from rocks mainly by which process? [\[Soil\]](#)

- (A) Weathering, as sun, water and wind slowly break rocks apart
- (B) Earthquakes alone shattering rocks in seconds
- (C) Photosynthesis by green plants
- (D) Burning of the rock by the Sun
- (E) Boiling of underground water
- (F) Magnetism pulling rock apart
- (G) Animals digging tunnels only

36. Among 0.7, 0.07, 0.77 and 0.707, which is the largest? [\[A Peek Beyond the Point\]](#)

- (A) 0.7
- (B) they are all equal
- (C) 0.077
- (D) 0.7007
- (E) 0.07
- (F) 0.77

37. White sunlight passes through a glass prism and spreads into a band of colours. Which colour bends the most and which bends the least? [\[Light\]](#)

- (A) Violet bends most, green bends least
- (B) Yellow bends most, blue bends least
- (C) Violet bends most, red bends least
- (D) Red bends most, violet bends least
- (E) Red bends most, green bends least
- (F) Green bends most, red bends least
- (G) All colours bend equally

38. Subtracting 4 from 3 times a number gives the same result as the number increased by 10. Find the number. [\[Simple Equations\]](#)

- (A) 5
- (B) 6
- (C) 7
- (D) 14
- (E) 3.5
- (F) 4.67

39. The dark, uppermost layer of soil that is rich in humus and supports plant growth is called what? [\[Soil\]](#)

- (A) Topsoil (the A-horizon)
- (B) The deepest mineral layer
- (C) Pure gravel layer
- (D) The water table
- (E) Parent rock
- (F) Bedrock
- (G) C-horizon

40. A bottle holds 1.5 L and a glass holds 0.125 L. How many full glasses can be filled from one bottle? [\[A](#)

[Peek Beyond the Point\]](#)

- (A) 15
- (B) 6
- (C) 1.2
- (D) 120
- (E) 12
- (F) 8

41. In the rainbow sequence remembered as VIBGYOR, counting from the violet end, which colour is the fifth? [\[Light\]](#)

- (A) Green
- (B) Yellow
- (C) Indigo
- (D) Red
- (E) Blue
- (F) Orange

42. Multiplying a number by 6 gives the same result as adding 40 to that number. Find the number. [\[Simple Equations\]](#)

- (A) 10
- (B) 8
- (C) 48
- (D) 5
- (E) 40
- (F) 7

43. If a soil's percolation rate is 15 mL/min, how long will 180 mL of water take to pass through it? [\[Soil\]](#)

- (A) 12 minutes
- (B) 0.083 minutes
- (C) 6 minutes
- (D) 2700 minutes
- (E) 120 minutes
- (F) 15 minutes

44. Riya buys items costing 45.75, 12.50 and 8.05 rupees and pays with a 100-rupee note. How much change does she get? [\[A Peek Beyond the Point\]](#)

- (A) 32.70 rupees
- (B) 43.70 rupees
- (C) 66.30 rupees
- (D) 33.30 rupees
- (E) 33.70 rupees
- (F) 34.70 rupees
- (G) 23.70 rupees

45. A prism produces the seven spectral colours. Which single colour lies exactly in the middle of the band? [\[Light\]](#)

- (A) Yellow
- (B) Blue
- (C) Orange
- (D) Red
- (E) Indigo
- (F) Green

46. Three notebooks together with 20 rupees cost the same as 5 such notebooks. Find the cost of one notebook. [\[Simple Equations\]](#)

- (A) 2 rupees
- (B) 8 rupees
- (C) 20 rupees
- (D) 40 rupees
- (E) 10 rupees
- (F) 5 rupees
- (G) 4 rupees

47. A field absorbs rainwater at 4 mm per hour. After a 3-hour shower that drops 18 mm of rain, how much water could not soak in and ran off? [\[Soil\]](#)

- (A) 12 mm
- (B) 2 mm
- (C) 22 mm
- (D) 14 mm
- (E) 18 mm
- (F) 6 mm

48. What is 0.25 multiplied by 0.25? [\[A Peek Beyond the Point\]](#)

- (A) 0.05
- (B) 0.625
- (C) 0.0625
- (D) 0.025
- (E) 0.125
- (F) 6.25
- (G) 0.5

49. A plane mirror reflects about 0.9 of the light that falls on it. If 250 units of light strike it, how many units are reflected and how many are absorbed? [\[Light\]](#)

- (A) 250 reflected, 0 absorbed
- (B) 180 reflected, 70 absorbed
- (C) 90 reflected, 160 absorbed
- (D) 200 reflected, 50 absorbed
- (E) 25 reflected, 225 absorbed
- (F) 225 reflected, 25 absorbed

50. The sum of two numbers is 50, and one number is 8 more than the other. Find the smaller number.

[Simple Equations]

- (A) 21
- (B) 23
- (C) 29
- (D) 42
- (E) 17
- (F) 25

51. Earthworms are called a farmer's friend mainly because they do what to the soil? [Soil]

- (A) Burrow through it, loosening it and letting in air and water
- (B) Make the soil waterproof
- (C) Kill the roots of plants
- (D) Turn the soil back into solid rock
- (E) Add harmful salt to the soil
- (F) Remove all the humus from soil

52. Express the decimal 0.045 as a fraction in its lowest terms. [A Peek Beyond the Point]

- (A) $\frac{9}{2000}$
- (B) $\frac{45}{100}$
- (C) $\frac{45}{10}$
- (D) $\frac{3}{200}$
- (E) $\frac{9}{200}$
- (F) $\frac{1}{45}$
- (G) $\frac{9}{20}$

53. Light keeps only 0.8 of its strength at each reflection. If 500 units enter a device and reflect twice, how much light remains? [Light]

- (A) 320 units
- (B) 360 units
- (C) 300 units
- (D) 256 units
- (E) 250 units
- (F) 400 units

54. A number is divided by 4 and then 7 is added, giving 19. Find the number. [Simple Equations]

- (A) 12
- (B) 44
- (C) 48
- (D) 104
- (E) 96
- (F) 3
- (G) 52

55. Which soil is the worst choice for a strong house foundation because it swells when wet and shrinks when dry? [\[Soil\]](#)

- (A) Coarse sand
- (B) Sandy soil
- (C) Rocky gravel soil
- (D) Solid bedrock
- (E) Pebbly soil
- (F) Loamy soil
- (G) Clayey soil

56. Work out 6.25 divided by 2.5, then add 1.5 to the answer. [\[A Peek Beyond the Point\]](#)

- (A) 5.5
- (B) 3.75
- (C) 2.0
- (D) 4.0
- (E) 4.25
- (F) 2.5

57. A wall clock seen in a plane mirror appears to show 4:40. What is the actual time? [\[Light\]](#)

- (A) 8:20
- (B) 7:20
- (C) 11:40
- (D) 5:20
- (E) 6:40
- (F) 4:40

58. In a class the number of girls is 5 more than twice the number of boys, and there are 35 students in all. How many boys are there? [\[Simple Equations\]](#)

- (A) 12
- (B) 15
- (C) 11
- (D) 8
- (E) 25
- (F) 10
- (G) 20

59. A soil mixture is 40 percent sand, 25 percent clay, and the rest silt by weight. In 200 kg of this soil, how many kilograms are silt? [\[Soil\]](#)

- (A) 70 kg
- (B) 50 kg
- (C) 130 kg
- (D) 80 kg
- (E) 45 kg
- (F) 65 kg

60. A car runs 0.8 km on each unit of fuel. How far will it travel on 12.5 units of fuel? [\[A Peek Beyond the Point\]](#)

- (A) 8 km
- (B) 12.5 km
- (C) 13.3 km
- (D) 10 km
- (E) 15.625 km
- (F) 1 km

61. The reflection of a clock in a mirror reads 7:30. What is the real time on the clock? [\[Light\]](#)

- (A) 4:00
- (B) 4:30
- (C) 5:30
- (D) 6:30
- (E) 11:30
- (F) 7:30
- (G) 8:30

62. Seven less than four times a number equals five times the number minus twelve. Find the number.

[\[Simple Equations\]](#)

- (A) 3
- (B) 12
- (C) 19
- (D) -5
- (E) 1
- (F) 7
- (G) 5

63. Soil pollution is caused mainly by which of the following? [\[Soil\]](#)

- (A) Ordinary rainwater soaking in
- (B) Adding natural humus to soil
- (C) Sunlight falling on the field
- (D) Earthworms living in the soil
- (E) Planting too many trees
- (F) Plastics, harmful chemicals and excess fertilisers or pesticides

64. Round the number 8.4651 to two decimal places. [\[A Peek Beyond the Point\]](#)

- (A) 8.47
- (B) 8.465
- (C) 8.46
- (D) 8.4
- (E) 8.466
- (F) 8.45
- (G) 8.50

- 65.** An ambulance has a word painted on its bonnet so it reads correctly in the mirrors of cars ahead. Why must the word be printed reversed? [\[Light\]](#)
- (A) Because the image is larger than the object
 - (B) Because light bends as it enters the mirror
 - (C) Because the image forms in front of the mirror
 - (D) Because a plane mirror reverses an image left-to-right (lateral inversion)
 - (E) Because a mirror flips images upside-down
 - (F) Because a mirror changes the colour of letters
- 66.** A rope is cut into two pieces, one 3 times as long as the other, and the longer piece is 90 cm. What is the total length of the rope? [\[Simple Equations\]](#)
- (A) 120 cm
 - (B) 360 cm
 - (C) 270 cm
 - (D) 60 cm
 - (E) 150 cm
 - (F) 90 cm
 - (G) 30 cm
- 67.** Soil X lets 300 mL through in 24 minutes; soil Y lets 150 mL through in 10 minutes. Which soil has the greater percolation rate? [\[Soil\]](#)
- (A) Both equal at 12.5 mL/min
 - (B) Soil Y, at 10 mL/min
 - (C) Soil X, at 24 mL/min
 - (D) Soil Y, at 15 mL/min
 - (E) Soil X, at 12.5 mL/min
 - (F) Soil Y, at 12.5 mL/min
- 68.** Subtract 0.999 from 5. What is the difference? [\[A Peek Beyond the Point\]](#)
- (A) 4.01
 - (B) 4.111
 - (C) 5.999
 - (D) 3.001
 - (E) 4.991
 - (F) 4.001
 - (G) 4.9
- 69.** Which set of properties correctly describes the image formed by a plane mirror? [\[Light\]](#)
- (A) Virtual, erect, same size, laterally inverted
 - (B) Virtual, inverted, larger
 - (C) Virtual, inverted, same size
 - (D) Real, inverted, smaller
 - (E) Virtual, erect, magnified
 - (F) Real, erect, laterally inverted
 - (G) Real, erect, same size

70. When 9 is added to a number and the sum is then multiplied by 2, the result is 50. Find the number.

[Simple Equations]

- (A) 41
- (B) 32
- (C) 16
- (D) 25
- (E) 20
- (F) 34

71. The layer just below the topsoil, harder and lower in humus but holding minerals washed down from above, is the? [Soil]

- (A) The water table
- (B) The deepest parent rock
- (C) Bedrock
- (D) The surface litter layer
- (E) Subsoil (the B-horizon)
- (F) Topsoil (A-horizon)
- (G) Loose humus layer

72. Multiply 0.012 by 1000. What is the product? [A Peek Beyond the Point]

- (A) 12
- (B) 0.12
- (C) 1.2
- (D) 120
- (E) 12000
- (F) 1200
- (G) 0.012

73. An object is 1.2 m from a plane mirror. The mirror is then moved 0.3 m further from the object. By how much does the image move away from the object? [Light]

- (A) 2.4 m
- (B) 0.3 m
- (C) 0.15 m
- (D) 0.6 m
- (E) 1.5 m
- (F) 1.2 m
- (G) 0.9 m

74. A square of side 9 cm and an equilateral triangle happen to share exactly the same perimeter. How long is each side of that triangle? [Simple Equations]

- (A) 12 cm
- (B) 27 cm
- (C) 36 cm
- (D) 9 cm
- (E) 4 cm
- (F) 18 cm
- (G) 6 cm

75. Why is loamy soil considered ideal for growing most plants? [\[Soil\]](#)

- (A) It is made of pure sand only
- (B) It holds enough water and air and is rich in humus and nutrients
- (C) It is mostly large stones
- (D) It never lets any water pass through
- (E) It is made of pure clay only
- (F) It contains no humus at all
- (G) It drains instantly like loose gravel

76. A wire 2.5 m long weighs 0.4 kg for every metre. What is the total weight of the wire? [\[A Peek Beyond the Point\]](#)

- (A) 10 kg
- (B) 2.9 kg
- (C) 6.25 kg
- (D) 2.5 kg
- (E) 1.0 kg
- (F) 0.16 kg
- (G) 0.625 kg

77. Two plane mirrors are set at an angle so that exactly 5 images of an object are formed. What is the angle between the mirrors? [\[Light\]](#)

- (A) 45 degrees
- (B) 30 degrees
- (C) 90 degrees
- (D) 120 degrees
- (E) 72 degrees
- (F) 36 degrees
- (G) 60 degrees

78. Half of a number is 6 more than one-third of the same number. Find the number. [\[Simple Equations\]](#)

- (A) 36
- (B) 12
- (C) 6
- (D) 18
- (E) 24
- (F) 72

79. A farmer wants a soil's percolation rate in litres per hour. The soil drains 0.5 L in 6 minutes. What is the rate per hour? [\[Soil\]](#)

- (A) 0.5 litres per hour
- (B) 0.083 litres per hour
- (C) 5 litres per hour
- (D) 60 litres per hour
- (E) 12 litres per hour
- (F) 3 litres per hour

80. Find the average of the three decimals 2.5, 3.5 and 4.8. [\[A Peek Beyond the Point\]](#)

- (A) 3.8
- (B) 3.6
- (C) 10.8
- (D) 3.5
- (E) 2.7
- (F) 5.4
- (G) 3.4

81. A small object is placed between two parallel plane mirrors facing each other. How many images are formed in theory? [\[Light\]](#)

- (A) Infinitely many
- (B) 2
- (C) 1
- (D) 7
- (E) 5
- (F) 0
- (G) 3

82. A taxi charges a fixed 30 rupees plus 12 rupees for each kilometre. A trip cost 102 rupees. How many kilometres was the trip? [\[Simple Equations\]](#)

- (A) 5 km
- (B) 72 km
- (C) 7 km
- (D) 9 km
- (E) 8.5 km
- (F) 6 km
- (G) 11 km

83. Which sequence correctly shows how soil forms over a very long time, from start to finish? [\[Soil\]](#)

- (A) Humus first, then rock forms on top of it
- (B) Plants make rock, which becomes soil
- (C) Bare rock, then weathering breaks it, then it mixes with humus, giving fertile soil
- (D) Soil forms in a single day from rain
- (E) Sand turns into rock, then into bedrock only
- (F) Bedrock turns straight into humus with no weathering
- (G) Fertile soil first, then it hardens into bare rock

84. Which is greater, 0.45 or $\frac{3}{8}$, and by how much? [\[A Peek Beyond the Point\]](#)

- (A) 0.45, by 0.075
- (B) 0.45, by 0.025
- (C) they are equal
- (D) $\frac{3}{8}$, by 0.125
- (E) $\frac{3}{8}$, by 0.075
- (F) 0.45, by 0.75
- (G) 0.45, by 0.05

85. In a pinhole camera, an object 20 cm tall stands 100 cm from the hole, and the screen is 25 cm behind the hole. How tall is the image on the screen? [\[Light\]](#)

- (A) 80 cm
- (B) 8 cm
- (C) 5 cm
- (D) 2.5 cm
- (E) 20 cm
- (F) 10 cm

86. Subtracting 15 from a number gives the same value as one-fourth of that number. Find the number.

[\[Simple Equations\]](#)

- (A) 12
- (B) 60
- (C) 20
- (D) 15
- (E) 5
- (F) 80

87. After heavy rain, water stands for a long time on clayey soil. This happens because clayey soil has a...

[\[Soil\]](#)

- (A) Too much air space
- (B) No water-holding capacity at all
- (C) Very fast drainage speed
- (D) High percolation rate
- (E) Low percolation rate (poor drainage)
- (F) Very large particles
- (G) Coarse, gritty texture

88. A recipe needs 0.75 kg of flour per cake. How much flour is needed for 6 cakes, and how much is left from a 5 kg bag? [\[A Peek Beyond the Point\]](#)

- (A) Need 5.25 kg; none left
- (B) Need 4.5 kg; 4.5 kg left
- (C) Need 4 kg; 1 kg left
- (D) Need 4.5 kg; 1.5 kg left
- (E) Need 4.5 kg; 0.5 kg left
- (F) Need 45 kg; none left

89. Sunlight looks white but is really a mix of seven colours. If a disc painted with these seven colours is spun very fast, how does it appear? [\[Light\]](#)

- (A) Bright green
- (B) Grey with no colour ever
- (C) Seven separate rings
- (D) Whitish
- (E) Black
- (F) Only red
- (G) Pure violet

90. Two consecutive even numbers add up to 86. What is the larger of the two? [\[Simple Equations\]](#)

- (A) 46
- (B) 45
- (C) 48
- (D) 43
- (E) 42
- (F) 40
- (G) 44

91. In a percolation test a funnel passes 240 mL of water in 16 minutes. A second soil's rate is exactly half of this. What is the second soil's percolation rate? [\[Soil\]](#)

- (A) 30 mL/min
- (B) 6 mL/min
- (C) 8 mL/min
- (D) 120 mL/min
- (E) 7.5 mL/min
- (F) 3.75 mL/min
- (G) 15 mL/min

92. Evaluate $(1.2 \times 1.2) - 0.44$. [\[A Peek Beyond the Point\]](#)

- (A) 2.4
- (B) 0.96
- (C) 2.84
- (D) 1.0
- (E) 1.44
- (F) 1.88

93. A certain mirror always forms an erect and smaller image, no matter where the object is placed. What kind of mirror is it? [\[Light\]](#)

- (A) Both concave and plane
- (B) A glass prism
- (C) Plane mirror
- (D) Convex mirror
- (E) Concave mirror
- (F) Concave lens
- (G) Convex lens

94. Buying 4 pencils together with a 5-rupee eraser costs 29 rupees. Find the cost of one pencil. [\[Simple Equations\]](#)

- (A) 5 rupees
- (B) 4 rupees
- (C) 6 rupees
- (D) 9 rupees
- (E) 24 rupees
- (F) 8.5 rupees
- (G) 7.25 rupees

95. Which statement correctly explains why clayey soil feels sticky and holds water while sandy soil feels gritty and dries quickly? [\[Soil\]](#)

- (A) Clay has very small, tightly packed particles; sand has large particles with big air gaps
- (B) Clay has large particles; sand has tiny particles
- (C) Both have the same particle size
- (D) Sand is sticky because it has the smallest particles
- (E) Clay dries fastest because its gaps are largest
- (F) Clay has big air gaps; sand has none

96. A tank fills 0.6 m in depth every hour. How many hours will it take to fill to a depth of 4.5 m? [\[A Peek Beyond the Point\]](#)

- (A) 7.5 hours
- (B) 4.5 hours
- (C) 9 hours
- (D) 2.7 hours
- (E) 0.133 hours
- (F) 5.1 hours
- (G) 75 hours

97. A concave mirror is used as a shaving or make-up mirror because, when the face is held close, the image is... [\[Light\]](#)

- (A) Real and magnified
- (B) Erect and smaller
- (C) Erect and the same size
- (D) Inverted and smaller
- (E) Inverted and magnified
- (F) Inverted and the same size
- (G) Erect, virtual and magnified

98. Three times a number reduced by 5 equals the number added to 21. Find the number. [\[Simple Equations\]](#)

- (A) 18
- (B) 16
- (C) 13
- (D) 6.5
- (E) 8
- (F) 26
- (G) 11

99. A 100 mL block of dry soil is found to contain 28 mL of empty pore space between its particles. If this pore space were instead 0.4 of the total volume, by how many mL would the air space increase? [\[Soil\]](#)

- (A) 72 mL
- (B) 40 mL
- (C) 60 mL
- (D) 32 mL
- (E) 28 mL
- (F) 12 mL

100. Using the correct order of operations, what is $0.2 + 0.2 \times 0.2$? [\[A Peek Beyond the Point\]](#)

- (A) 0.06
- (B) 0.16
- (C) 0.4
- (D) 0.08
- (E) 0.24
- (F) 0.44